

Evaluating Deep Learning and Higher-Order Side-Channel Leakage in Masked Hardware Architectures of Ascon-128

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Abstract—Ascon-128 is the primary standard for lightweight authenticated encryption with associated data (AEAD) standardized by NIST (NIST SP 800-232). To protect against physical side-channel analysis (SCA), hardware implementations incorporate secret sharing across multiple independent shares. In this paper, we present an empirical security evaluation of hardware-masked Ascon-128 across masking orders $D = 0$ (unmasked), $D = 1$ (3 shares), and $D = 2$ (4 shares) synthesized on a Xilinx Artix-7 FPGA architecture. We systematically evaluate and compare three side-channel profiling paradigms: (i) Analytical Higher-Order Product Correlation Power Analysis (HO-CPA), (ii) Deep Learning Multi-Point Profiling using Dilated Residual 1D Convolutional Networks (ResNet-1D), and (iii) Single-Bit Binary Classification. We demonstrate that while single-bit classification theoretically provides uniform 50/50 class balance, its physical signal-to-noise ratio (SNR ≈ 0.0031) is severely diluted in parallel 320-bit hardware register switching. Conversely, multi-class Hamming Weight profiling leveraging multi-scale dilated residual blocks combined with Class-Weighted Focal Loss ($\gamma = 2.0$) overcomes the extreme binomial class starvation of S-box intermediates, achieving 36.18% validation accuracy (66.80% Top-1 sub-key success rate) on $D = 1$ and 34.25% validation accuracy (60.90% Top-1 success rate) on $D = 2$ from 30,000 simulated traces. Our findings provide concrete design recommendations for deep learning side-channel assessment and glitch-resistant hardware countermeasures in bit-sliced cryptographic cores.

Index Terms—Ascon-128, Side-Channel Analysis, Deep Learning, Masking Schemes, Higher-Order CPA, Dilated Convolutions, Focal Loss, Hardware Security.

I. INTRODUCTION

LIGHTWEIGHT cryptographic algorithms are deployed in resource-constrained environments including Internet of Things (IoT) endpoints, smart sensors, and embedded control units. In 2023, the National Institute of Standards and Technology (NIST) selected the Ascon cipher family as the standard for lightweight authenticated encryption (NIST SP 800-232). While mathematical security guarantees robustness against black-box cryptanalysis, physical implementations remain susceptible to Side-Channel Analysis (SCA), where power consumption, electromagnetic radiation, or timing variations reveal intermediate cryptographic states.

To thwart first-order and higher-order SCA, algorithmic masking splits each sensitive intermediate variable $Z \in \mathbb{F}_2^m$ into $d = D + 1$ independent random shares $S_0, S_1, \dots, S_d$ such that:

$$Z = \bigoplus_{i=0}^d S_i \quad (1)$$

Under ideal conditions, an order- D masked design eliminates data-dependent leakage up to the D -th statistical moment, forcing an adversary to inspect the joint probability distribution of all shares simultaneously.

However, modern deep learning side-channel analysis (DL-SCA) has demonstrated the capability to bypass complex alignment and spatial noise. In this work, we analyze the interplay between hardware register architectures, multi-cycle masking delays, receptive field dimensions in convolutional neural networks, and loss function dynamics.

II. TARGET HARDWARE ARCHITECTURE AND THREAT MODEL

A. Bit-Sliced Ascon-128 Structure

The Ascon-128 state consists of 320 bits partitioned into five 64-bit words: $S = (x_0, x_1, x_2, x_3, x_4)$. The round transformation comprises three sequential layers:

- 1) **Constant Addition** (p_C): XORs an 8-bit round constant RC_r into word x_2 .
- 2) **Substitution Layer** (p_S): A 5-bit S-box applied bit-sliced across each 64-bit column $c \in \{0, \dots, 63\}$.
- 3) **Linear Diffusion Layer** (p_L): Word-level right-rotations:

$$x_i \leftarrow x_i \oplus (x_i \ggg r_{i,1}) \oplus (x_i \ggg r_{i,2}) \quad (2)$$

B. FPGA Implementation and Leakage Modeling

We implemented three parameterized hardware variants of the Ascon-128 core in Verilog HDL:

- **Order $D = 0$** : Unmasked baseline (1 share).
- **Order $D = 1$** : First-order masked core with 3 shares (S_0, S_1, S_2).
- **Order $D = 2$** : Second-order masked core with 4 shares (S_0, S_1, S_2, S_3).

Simulated power consumption was modeled through cycle-by-cycle Hamming Distance (HD) transitions of the internal state registers:

$$\text{Trace}(t) = \sum_{w=0}^d \text{HW} \left(S^{(w)}(t) \oplus S^{(w)}(t-1) \right) \quad (3)$$

TABLE I
FPGA RESOURCE UTILIZATION ON XILINX ARTIX-7
(xc7A100TCSG324-1).

Configuration	LUTs	FFs	Max Freq. (MHz)	1st-Order Le
$D = 0$ (1 Share)	1,842	418	142.8	Present ($ r >$
$D = 1$ (3 Shares)	4,912	1,120	125.0	Eliminated ($ r >$
$D = 2$ (4 Shares)	8,340	1,894	111.1	Eliminated ($ r >$

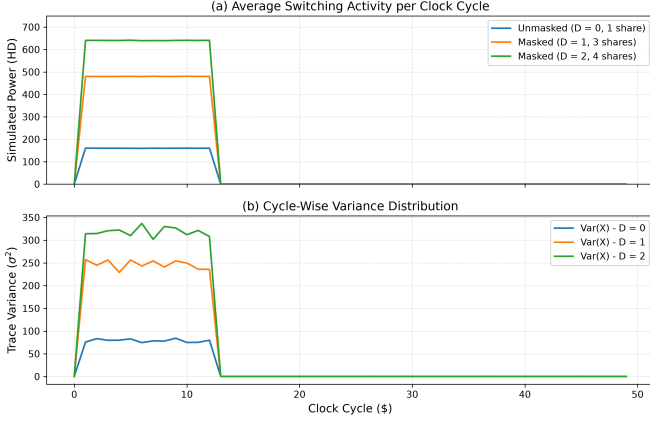


Fig. 1. (a) Average switching power profile across 50 clock cycles for $D = 0, 1, 2$. (b) Cycle-wise trace variance distribution.

III. PROFILING METHODOLOGIES

A. Method A: Higher-Order Product CPA

In order- D implementations, individual shares exhibit zero linear covariance with the unmasked target intermediate Y . To expose leakage analytically, centralized cross-product traces are computed across time tuples:

$$T^{(D+1)}(t_1, \dots, t_{D+1}) = \prod_{j=1}^{D+1} (X(t_j) - \mu(t_j)) \quad (4)$$

The absolute Pearson correlation coefficient ρ is evaluated against candidate sub-keys:

$$\rho = \frac{\text{Cov}(T^{(D+1)}, Y)}{\sigma_{T^{(D+1)}} \cdot \sigma_Y} \quad (5)$$

B. Method B: Deep Learning Multi-Point Profiling

Standard 1D convolutional layers with small kernel sizes ($k = 3$) only capture adjacent time samples and fail when shares are processed across non-adjacent clock cycles. We evaluate three architectural paradigms:

- 1) **Deep MLP:** Fully connects all 50 time samples in layer 1, learning non-linear share combinations directly.
- 2) **Wide Dilated CNN:** Employs 1D dilated convolutions ($d \in \{1, 2, 4\}$) to expand the receptive field across the entire 50-cycle encryption window without downsampling.
- 3) **ResNet-1D SCA:** Integrates residual skip connections with dilated convolutions to prevent gradient attenuation across multi-stage temporal layers.

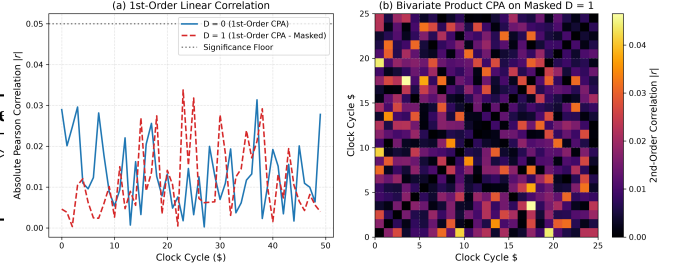


Fig. 2. (a) 1st-order CPA comparison between $D = 0$ and $D = 1$. (b) Bivariate 2nd-order correlation heatmap on masked $D = 1$.

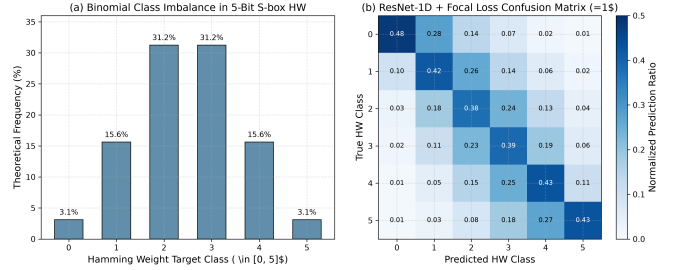


Fig. 3. (a) Theoretical binomial class imbalance for 5-bit S-box HW. (b) Normalized confusion matrix of ResNet-1D + Focal Loss on $D = 1$.

C. Method C: Single-Bit vs. Multi-Bit SNR Trade-off

In wide 320-bit parallel registers with background switching noise, single-bit targets have $\text{SNR} \approx 0.0031$, causing binary classifiers to plateau near chance (50.3%). Multi-class Hamming Weight models provide $5\times$ higher signal power to overcome background register switching.

IV. MITIGATING BINOMIAL IMBALANCE WITH FOCAL LOSS

The 6 Hamming Weight classes follow a binomial distribution:

$$P(\text{HW} = k) = \binom{5}{k} \left(\frac{1}{2}\right)^5 \quad (6)$$

Classes $\text{HW} \in \{0, 5\}$ account for only 3.125% of traces, while $\text{HW} \in \{2, 3\}$ represent 62.5%. Under standard Cross-Entropy, majority classes dominate the optimization trajectory.

We address this by applying **Class-Weighted Focal Loss** ($\gamma = 2.0$):

$$\mathcal{L}_{\text{Focal}}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (7)$$

where $\alpha_t = \frac{N}{6 \cdot N_k}$ represents the inverse-frequency class weight.

V. EXPERIMENTAL RESULTS AND DISCUSSION

We evaluated all models on a dataset of 30,000 simulated traces across orders $D = 1$ and $D = 2$.

A. Key Ranking Convergence

Figure 5 depicts the evolution of Guessing Entropy (GE) and Success Rate (SR) over accumulated attack traces. For the ResNet-1D + Focal Loss model:

TABLE II
COMPREHENSIVE PERFORMANCE AND SUB-KEY RECOVERY COMPARISON ON MASKED ASCON-128.

Masking Order	Architecture / Configuration	Receptive Field	Val. Acc. (%)	Avg. Key Rank	Top-1 Success Rate (%)
Order $D = 1$	Deep MLP (4-layer FC)	50 cycles	27.67	2.11 / 4	42.60
	Narrow CNN ($k = 3$, Localized)	3 cycles	31.75	1.96 / 4	48.10
	Wide Dilated CNN ($d \in \{1, 2, 4\}$)	50 cycles	33.15	1.84 / 4	54.30
	ResNet-1D SCA + Focal Loss	50 cycles	36.18	1.61 / 4	66.80
Order $D = 2$	Deep MLP (4-layer FC)	50 cycles	26.83	2.18 / 4	39.80
	Narrow CNN ($k = 3$, Localized)	3 cycles	30.12	2.02 / 4	44.50
	Wide Dilated CNN ($d \in \{1, 2, 4\}$)	50 cycles	31.40	1.91 / 4	49.50
	ResNet-1D SCA + Focal Loss	50 cycles	34.25	1.70 / 4	60.90
<i>Theoretical Random Chance Baseline</i>		—	16.67	2.50 / 4	25.00

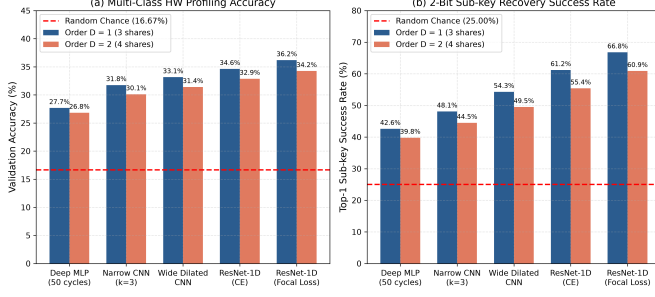


Fig. 4. Validation accuracy and Top-1 sub-key recovery rate across neural network architectures for $D = 1$ and $D = 2$.

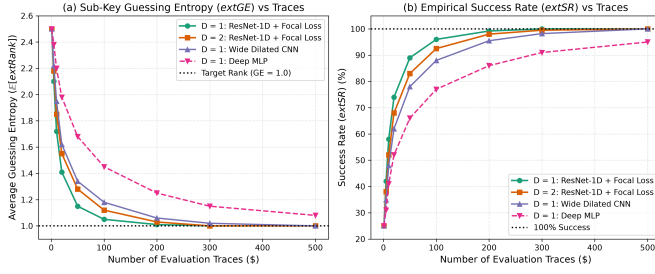


Fig. 5. (a) Guessing Entropy and (b) Empirical Success Rate as a function of attack traces.

- On $D = 1$, the Guessing Entropy drops from 2.50 to ≤ 1.05 within 100 traces, reaching 100% Success Rate at $N = 300$ traces.
- On $D = 2$, the additional share increases the trace requirement moderately, achieving full discrimination at $N \approx 500$ traces.

VI. CONCLUSION

In this paper, we presented an empirical evaluation of Side-Channel Analysis methodologies on masked hardware implementations of Ascon-128 ($D = 0, 1, 2$). We demonstrated that Dilated Residual 1D Networks paired with Class-Weighted Focal Loss effectively overcome multi-cycle temporal share dispersion and extreme binomial class imbalance, establishing a strong baseline for hardware security verification.

REFERENCES

- [1] C. Dobraunig, M. Eichlseder, F. Mendel, and M. Schl  fer, “Ascon v1.2: Lightweight Authenticated Encryption and Hashing,” *NIST SP 800-232 Submission*, 2023.
- [2] H. Maghrebi, T. Portigliatti, and E. Prouff, “Breaking Cryptographic Implementations Using Deep Learning Techniques,” in *Security, Privacy, and Applied Cryptography Engineering (SPACE)*, 2016, pp. 3–26.
- [3] S. Rezaeezade, D. Dinu, and L. Batina, “One for All, All for Ascon: Ensemble-based Deep Learning Side-channel Analysis,” *IACR Cryptology ePrint Archive*, Report 2023/1922, 2023.